Lectures On Class Field Theory

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Contents

1 Un peu d'histoire **2**
1.1 Reciprocity laws 2
1.2 Kummer case 3

2 Computing the Hilbert symbol **5**

3 Cohomology of groups **8**
3.1 Restriction, Inflation and Corestriction 9
3.2 Profinite groups 9
3.3 Cohomology of profinite groups 9

4 Kummer Theory **11**

5 Cup-products **13**

1 Un peu d'histoire

We start with some motivation: this course will be about number fields (i.e. finite extensions $K|\mathbf{Q}$) and their absolute Galois groups $G_K = \mathrm{Gal}(\overline{K}|K)$, where $\overline{K}$ is a fixed algebraic closure of K. Very little is known about the structure of such automorphism groups in general, and a large chunk of modern number theory/arithmetic geometry is devoted to studying them, e.g. via their representations. In some sense, class field theory consists in the study of its 1-dimensional representations; more precisely, it describes the abelianization $G_K^{\mathrm{ab}} = G_K/[G_K, G_K]$ of the absolute Galois group (equivalently, its abelian quotients).

1.1 Reciprocity laws

Let p be a prime number, and let a be an integer coprime to p. The *Legendre symbol* $\left(\frac{a}{p}\right)$ is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square modulo } p, \\ -1 & \text{if } a \text{ is not a square modulo } p. \end{cases}$$

Euler proved that $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p$, and **Gauss** proved the following law:

Theorem 1.1 (Quadratic reciprocity)**.** Let $a \neq b$ be different odd primes. Then,

$$\left(\frac{a}{b}\right) \cdot \left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}.$$

Morever, the following subsidiary laws hold:

1. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$;
2. $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$.

These laws, together with its multiplicativity in the parameter a, make it possible to compute the Legendre symbol.

Another way of thinking about class field theory is as an attempt to define a "very general" reciprocity law, from which the quadratic reciprocity law and its higher degree analogues can be deduced. One such generalization is **Hilbert**'s reciprocity law: recall that $\mathbf{Q}$ embeds into the p-adic field $\mathbf{Q}_p$ for every prime p, and also into the real numbers $\mathbf{Q}_\infty \coloneqq \mathbf{R}$. For every prime p (including $p = \infty$) and for all $a, b \in \mathbf{Q}^\times$, we define the *Hilbert symbol* $(a, b)_p$ as

$$(a, b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ has a solution in } \mathbf{Q}_p^2, \\ -1 & \text{otherwise.} \end{cases} \quad asd$$

Recall that the norm $N : \mathbf{Q}_p(\sqrt{a}) \to \mathbf{Q}_p$ is defined as $N(x + y\sqrt{a}) = x^2 - ay^2$ if $a \notin \mathbf{Q}_p^{\times 2}$, and as $N = \mathrm{id}$ otherwise. More generally, if $L|K$ is a finite Galois extension with group G, the norm $N_{L|K} : L \to K$ is

$$N(x) = \prod_{\sigma \in G} \sigma(x)$$

Lemma 1.2. The following statements are equivalent:

1. $(a,b)_p = 1$;
2. The quadratic equation $z^2 = ax^2 + by^2$ has a non-trivial solution;
3. $a \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{b})$.
4. $b \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{a})$.

Proof. The equivalence $1 \iff 2$ is by definition; and $3 \iff 4$ is clear. For $2 \iff 3$, if $b \in \mathbf{Q}_p^{\times 2}$, then we are done. Otherwise, if $b \notin \mathbf{Q}_p^{\times 2}$, then $N(\alpha + \beta\sqrt{b}) = \alpha^2 - b\beta^2$, and if $a \coloneqq \alpha^2 - b\beta^2$ we see that $z^2 = ax^2 + by^2$ has a non-trivial solution, and vice versa. □

For $p \neq 2, \infty$ and $a, b \in \mathbf{Z}$, write $a = p^{v_p(a)}u$ and $b = p^{v_p(b)}v$ with $p \nmid uv$. The relation between the Hilbert symbol and the Legendre symbol is

$$(a,b)_p = (-1)^{\frac{p-1}{2}v_p(a)v_p(b)} \left(\frac{u}{p}\right)^{v_p(b)} \left(\frac{v}{p}\right)^{v_p(a)} \tag{1.1}$$

$$= \overline{(-1)^{v_p(a)v_p(b)} a^{v_p(b)} b^{-v_p(a)}}^{(p-1)/2}. \tag{1.2}$$

where the bar denotes reduction modulo p. In particular, if p does not divide a, then $(a,b)_p = \left(\frac{a}{p}\right)^{v_p(b)}$ coincides with the Legendre symbol.

Theorem 1.3 (Hilbert reciprocity law)**.** For all $a, b \in \mathbf{Q}^\times$, we have

$$\prod_p (a,b)_p = 1$$

As for computation, one shows that if p does not divide $2ab$ (including $p = \infty$ for $a, b > 0$), then $(a,b)_p = 1$. Moreover, $(a,b)_2 = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}$ for odd a, b. From this and multiplicativity in the first variable, we immediately deduce the quadratic reciprocity law by taking a, b to be odd primes.

An advantage of this formulation is that it generalizes to number fields K, by considering the completions of K at all of its (finite and infinite) "places". Hilbert then went on to pose his 9th problem, in which he asked for a generalization of this law to n-th powers in arbitrary number fields; this is what class field theory achieves. We first consider the case of extensions which contain a primitive n-th root of unity:

1.2 Kummer case

Assume now $K|\mathbf{Q}$ is a finite extension such that $\mu_n \subset K$, and let $\mathcal{O}_K$ be its ring of integers. If $P \subset \mathcal{O}_K$ is a prime ideal, we would like to define a symbol $(a,b)_P \in \mu_n$ such that a product formula holds, i.e.

$$\prod_P (a,b)_P = 1$$

if $a, b \in K^\times$. Moreover, for P finite (which is always the case for $n \neq 2$) which does not divide (n) we shall have

$$(a,b)_P = \overline{(-1)^{v_P(a)v_P(b)} a^{v_P(b)} b^{-v_P(a)}}^{(q-1)/n},$$

where $q = \#(\mathcal{O}_K/P)$. In particular, if P does not divide (a), then $(a,b)_P = a^{\frac{q-1}{n} v_P(b)} \bmod P$.

So, if $v_P(b) = 1$, then

$$\left(\frac{a}{P}\right) \coloneqq (a,b)_P$$

does not depend on the choice of b and it is the unique n-th root of unity, that is $a^{\frac{q-1}{n}} \bmod P$; in other words, $\left(\frac{a}{P}\right) = 1$ if and only if $x^n = a \bmod P$ has a solution. Some more properties of this symbol are:

1. $(a,b)_P = 1 \iff b$ is a norm from $K_P(\sqrt[n]{a}) \iff a$ is a norm from $K_P(\sqrt[n]{b})$;
2. $(a,b)_P$ is multiplicative in both variables and anticommutative;

When μ_n is *not* contained in K_P, an "Hilbert" symbol still exists but is longer associated to $a, b \in K^\times$. Suppose for a moment that $\mu_n \subset K_P$. Then, the extension $K_P(\sqrt[n]{a})|K_P$ is Galois for all $a \notin K_P^{\times n}$, and its Galois group is

$$G_a \coloneqq \mathrm{Gal}(K_P(\sqrt[n]{a})|K_P) \cong \mathbf{Z}/n\mathbf{Z}$$

Then, the group $\mathrm{Hom}(G_a, \mu_n) \cong \mathbf{Z}/n\mathbf{Z}$ is generated by the character

$$\overline{\chi_a} : \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n \cong \mathbf{Z}/n\mathbf{Z}.$$

In general, we shall construct a generalized Hilbert symbol

$$\begin{aligned} \mathrm{Hom}\left(\mathrm{Gal}(\overline{K_P}|K_P), \mathbf{Z}/n\mathbf{Z}\right) \times K_P^\times &\to \mathbf{Z}/n\mathbf{Z}, \\ [\chi, b] &\mapsto (\chi, b)_P \end{aligned}$$

and we'll thus be able to prove the **Artin** reciprocity law: fix some $b \in K^\times$, and suppose that χ is a character in $\mathrm{Hom}(G_K, \mathbf{Z}/n\mathbf{Z})$. Then

$$\sum_P (\chi_P, b) = 0,$$

where χ_P is the restriction of χ to $G_{K_P}^{\mathrm{ab}}$ (recall that it embeds into G_K^{ab}).

The generalized Hilbert symbol defines a reciprocity map

$$K_P^\times \to G_{K_P}^{\mathrm{ab}},$$

which is "almost" an isomorphism. In particular, for every n we'll have an isomorphism of finite groups

$$K_P^\times/n \to G_{K_P}^{\mathrm{ab}}/n.$$

This is one of the remarkable features of class field theory, giving a description of the abelian extensions of a field in terms of its arithmetic. To define such a map, it's enough to define the dual map

$$\begin{aligned} \mathrm{Hom}(G_{K_P}^{\mathrm{ab}}/n, \mathbf{Z}/n\mathbf{Z}) &\to \mathrm{Hom}(K_P^\times/n, \mathbf{Z}/n\mathbf{Z}), \\ [\chi] &\mapsto [b \mapsto (\chi, b)_P]. \end{aligned}$$

2 Computing the Hilbert symbol

Recall the definition: for $a, b \in \mathbf{Q}_p^\times$,

$$(a,b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ is solvable in } \mathbf{Q}_p, \\ -1 & \text{otherwise.} \end{cases}$$

We have seen that

$(a,b)_p = 1 \iff z^2 - ax^2 - by^2$ has a non-trivial solution in $\mathbf{Q}_p^3 \iff b$ is a norm from $\mathbf{Q}_p(\sqrt{a})$,

where $N(x + y\sqrt{a}) = x^2 - ay^2$.

Theorem 2.1 (Hilbert reciprocity law)**.** For all $a, b \in \mathbf{Q}^\times$,

$$\prod_p (a,b)_p = 1,$$

where the product runs over all primes p, including $p = \infty$.

Note that the product is finite: $(a,b)_p = 1$ for all $p \nmid 2ab$ (see Lemma 2.4 below).

Lemma 2.2. Let $a, b, c \in \mathbf{Q}_p^\times$. Then

1. $(1,c)_p = 1$;
2. $(-c,c)_p = 1$;
3. if $(a,c)_p = 1$, then $(ab,c)_p = (a,c)_p(b,c)_p$ (this holds in general, see the Corollary below);
4. $(c,c)_p = (-1,c)_p$.

Proof. 1. Consider $N : \mathbf{Q}_p(\sqrt{c}) \to \mathbf{Q}_p$. Then $N(1) = 1$.

2. If $c \in \mathbf{Q}_p^{\times 2}$, then $\mathbf{Q}_p(\sqrt{c}) = \mathbf{Q}_p$, the norm is the identity and $N(-c) = -c$. If $c \notin \mathbf{Q}_p^{\times 2}$, then $N(\sqrt{c}) = -c$. In both cases $-c$ is a norm from $\mathbf{Q}_p(\sqrt{c})$.

3. If $a = N(x)$ and $b = N(y)$, then $ab = N(x)N(y) = N(xy)$, since the norm is multiplicative by definition.

   Moreover, if $a = N(x)$, then $a^{-1} = N(x^{-1})$; if in addition $ab = N(z)$, then

   $$b = a^{-1}(ab) = N(x^{-1})N(z) = N(x^{-1}z).$$

   So, if $(a,c)_p = 1$ (i.e. $a = N(x)$ is a norm from $\mathbf{Q}_p(\sqrt{c})$), then $(b,c)_p = 1 \iff (ab,c)_p = 1$.

4. By (2) and (3),

   $$(c,c)_p = (-c,c)_p(c,c)_p = (-c^2,c)_p = (-1,c)_p,$$

   because in general $(ad^2,b)_p = (a,b)_p$. □

Remark 2.3. Since $(ad^2, b)_p = (a, b)_p$, the Hilbert symbol is a well-defined map

$$(\,\cdot\,,\cdot\,)_p : \left(\mathbf{Q}_p^\times/\mathbf{Q}_p^{\times 2}\right) \times \left(\mathbf{Q}_p^\times/\mathbf{Q}_p^{\times 2}\right) \to \mu_2.$$

Lemma 2.4. Let $p > 2$. Then $(u, v)_p = 1$ for all $u, v \in \mathbf{Z}_p^\times$. (Recall that $\mathbf{Z}_p$ is local and $\mathbf{Z}_p^\times = \mathbf{Z}_p \setminus (p)$.)

Proof. Let $\overline{u} \coloneqq u \bmod p$ and $\overline{v} \coloneqq v \bmod p$; they are non-zero elements of $\mathbf{F}_p$ (units of the local ring). We first show that $1 - \overline{u}x^2 - \overline{v}y^2$ has a solution in $\mathbf{F}_p$. Since $\mathbf{F}_p^\times$ is cyclic of order $p-1$,

$$\#\{x^2 : x \in \mathbf{F}_p\} = \frac{p+1}{2}.$$

Similarly, if $S_{\overline{u}} \coloneqq \{\overline{u}x^2 : x \in \mathbf{F}_p\}$ and $S_{\overline{v}} \coloneqq \{1 - \overline{v}y^2 : y \in \mathbf{F}_p\}$, then $\#S_{\overline{u}} = \#S_{\overline{v}} = \frac{p+1}{2}$. Thus, $S_{\overline{u}} \cap S_{\overline{v}} \neq \emptyset$ by the

We conclude by Hensel's lemma: □

Lemma 2.5 (Hensel)**.** Let $f \in \mathbf{Z}_p[X]$ and $a_0 \in \mathbf{Z}_p$ with $f(a_0) \equiv 0 \mod p$ and $f'(a_0) \not\equiv 0 \mod p$. Then there exists $a \in \mathbf{Z}_p$ with $f(a) = 0$ and $a \equiv a_0 \mod p$.

This allows us to lift a solution mod p to a solution in $\mathbf{Z}_p$ (to apply it in the proof above, we have to fix one variable).

Proposition 2.6. Let $p > 2$ and $a, b \in \mathbf{Q}_p^\times$, with $a = p^\alpha u$, $b = p^\beta v$ and $u, v \in \mathbf{Z}_p^\times$. Then

$$(a, b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)^{-\beta} \left(\frac{\overline{v}}{p}\right)^{\alpha}.$$

Proof. Since $(a, b)_p$ depends only on $a, b \bmod \mathbf{Q}_p^{\times 2}$, we may assume $\alpha, \beta \in \{0, 1\}$.

- $\alpha = \beta = 0$: this is Lemma 2.4.
- $\alpha = 1$, $\beta = 0$: we need $(pu, v)_p = \left(\frac{\overline{v}}{p}\right)$. By Lemma 2.4 we have $(u^{-1}, v)_p = 1$, hence by Lemma 2.2(3)
$$(pu, v)_p = (pu, v)_p \underbrace{(u^{-1}, v)_p}_{=1} = (p, v)_p.$$
If $v \in \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = (p, 1)_p = 1 = \left(\frac{\overline{v}}{p}\right)$ (note that $\overline{v}$ is then also a square mod p). If $v \notin \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = -1$, because $z^2 - px^2 - vy^2$ has no non-trivial solution mod p, hence none in $\mathbf{Q}_p$ by homogeneity; also $\left(\frac{\overline{v}}{p}\right) = -1$ (by Hensel, since $v \notin \mathbf{Q}_p^{\times 2}$).

- $\alpha = \beta = 1$: we need $(pu, pv)_p = (-1)^{\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right)$. By Lemma 2.2(2) we have $(-pv, pv)_p = 1$, so by Lemma 2.2(3)

$$(pu, pv)_p = (pu, pv)_p \underbrace{(-pv, pv)_p}_{=1} = (-p^2uv, pv)_p = (-uv, pv)_p = (pv, -uv)_p,$$

and now we are in the previous case, which gives

$$(pv, -uv)_p = \left(\frac{\overline{-uv}}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right),$$

where $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$, and we used that the Legendre symbol is multiplicative. □

Corollary 2.7. Let $p > 2$. Then $(a, c)_p(b, c)_p = (ab, c)_p$ for all $a, b, c \in \mathbf{Q}_p^\times$. In particular, $(\cdot, \cdot)_p$ is bi-multiplicative.

Proof. Write $a = p^\alpha u$, $b = p^\beta v$ and $c = p^\gamma w$ with $u, v, w \in \mathbf{Z}_p^\times$. By Proposition 2.6,

$$\begin{aligned}(a, c)_p(b, c)_p &= (-1)^{\frac{p-1}{2}\alpha\gamma}\left(\frac{\overline{u}}{p}\right)^\gamma\left(\frac{\overline{w}}{p}\right)^\alpha \cdot (-1)^{\frac{p-1}{2}\beta\gamma}\left(\frac{\overline{v}}{p}\right)^\gamma\left(\frac{\overline{w}}{p}\right)^\beta \\ &= (-1)^{\frac{p-1}{2}\gamma(\alpha+\beta)}\left(\frac{\overline{uv}}{p}\right)^\gamma\left(\frac{\overline{w}}{p}\right)^{\alpha+\beta} = (ab, c)_p.\end{aligned}$$

□

Case $p = 2$. If $a = 2^\alpha u$ and $b = 2^\beta v$ with $u, v \in \mathbf{Z}_2^\times$, define

$$\varepsilon(u) \coloneqq \frac{u-1}{2} \bmod 2, \qquad \omega(u) \coloneqq \frac{u^2-1}{8} \bmod 2.$$

Then

$$(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(v)+\alpha\omega(v)+\beta\omega(u)}.$$

(Proof not given.) In particular, $(\cdot, \cdot)_p$ is bi-multiplicative for every p.

Proof of Theorem 2.1. Using bi-multiplicativity, we reduce to the case where each of a, b is either -1 or a prime.

- Case $a = -1$: homework.
- Case $a = b = q$ prime: $(q, q)_p = (-1, q)_p$ by Lemma 2.2(4), so we are reduced to the previous case.
- Case $a = 2$, $b > 2$ prime: $(2, b)_p = 1$ for $p \notin \{2, b\}$ (also for $p = \infty$, as $2, b > 0$). Moreover, by the $p = 2$ formula,

$$(2, b)_2 = (-1)^{\omega(b)},$$

and by Proposition 2.6

$$(2, b)_b = \left(\frac{2}{b}\right) = (-1)^{\frac{b^2-1}{8}}.$$

The product of these two factors is 1.

- Case $a \neq b$ odd primes: $(a,b)_p = 1$ for $p \notin \{2, a, b\}$ (as well as $p = \infty$), and

$$(a,b)_p = \begin{cases} (-1)^{\frac{a-1}{2}\frac{b-1}{2}} & p = 2, \\ \left(\frac{a}{b}\right) & p = b, \\ \left(\frac{b}{a}\right) & p = a. \end{cases}$$

  Hence the product is 1 by quadratic reciprocity. □

3 Cohomology of groups

Already for the local part of theory we'll require some input from homological algebra, mainly the description of the first cohomology group $H^1(G, A)$ for G some Galois group, and its interpretation in terms of crossed homomorphisms. A concise reference for this material can be found in the notes [2]. We recall some results here for the reader's convenience:

Theorem 3.1 (Cohomology of cyclic groups)**.** Let G be a finite cyclic group of order n, generated by σ. Then,

$$\begin{cases} H^0(G, A) = A^G, \\ H^1(G, A) = A/(1-\sigma)A, \\ H^2(G, A) = A^G/NA, \\ H^i(G, A) \cong H^{i+2}(G, A) \text{ for all } i \geq 1, \end{cases}$$

This comes out of the "periodic" resolution of $\mathbf{Z}$ as a trivial G-module:

$$\ldots \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\varepsilon} \mathbf{Z} \to 0,$$

where $N = 1 + \sigma + \ldots + \sigma^{n-1}$ is the norm map and ε sends σ to 1. A similar computation holds for the infinite cyclic group $G = \mathbf{Z}$:

$$H^0(\mathbf{Z}, A) = A^{\mathbf{Z}}, \qquad H^1(\mathbf{Z}, A) = A/(1-\sigma)A, \qquad H^i(\mathbf{Z}, A) = 0 \text{ for all } i \geq 2.$$

For a subgroup $H \subset G$, we denote by

$$M_H^G(A) \coloneqq \mathrm{Hom}_H(\mathbf{Z}[G], A)$$

the G-module with action

$$\sigma(\phi)(g) = \phi(\sigma^{-1}g), \qquad \sigma \in G, g \in G, \phi \in M_H^G(A).$$

Then, *Shapiro*'s lemma tells us that there is a canonical isomorphism

$$H^i(G, M_H^G(A)) \cong H^i(H, A). \tag{3.1}$$

3.1 Restriction, Inflation and Corestriction

Let H be a subgroup of G. The *restriction* map is defined on cocycles as

$$\mathrm{Res} : H^i(G, A) \to H^i(H, A), \quad \mathrm{Res}([f]) = [f|_H],$$

which explains the name: $f|_H$ is just restriction to H.

For a normal subgroup $H \triangleleft G$, the quotient G/H acts on A^H and we have the *inflation* map

$$\mathrm{Inf} : H^i(G/H, A^H) \to H^i(G, A), \quad \mathrm{Inf}([f]) = [f \circ \pi],$$

where $\pi : G \to G/H$ is the canonical projection.

The *corestriction* map goes in the opposite direction of restriction, and is defined as follows: let $H \subset G$ be a subgroup of finite index This is well-defined and gives a map

$$\mathrm{Cor} : H^i(H, A) \to H^i(G, A).$$

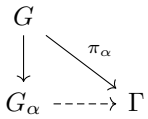
3.2 Profinite groups

Let $G = \varprojlim G_\alpha$ be a profinite group (that is, an inverse limit of finite groups). This is a topological group, whose topology is induced by the discrete topology on the finite groups G_α.

We have natural continuous maps $\pi_\alpha : G \to G_\alpha$, and their kernels U_α form a system of open neighborhoods of $1 \in G$.

Remark 3.2.

- We may assume that the G_α coincide with **all** finite quotients of G by open normal subgroups, as if Γ is such a quotient, then there is some α such that U_α has trivial image in Γ: indeed, Γ is finite and the intersection of all U_α is trivial. So, we get a factorization



- The inverse limit of a system of finite groups is compact and totally disconnected. With compact here we also include the Hausdorff property, which holds by Tychonoff's theorem. Moreover, the open subgroups of G are exactly the closed subgroups of finite index.

3.3 Cohomology of profinite groups

Let $G = \varprojlim G_\alpha$ be a profinite group, and let A be a *continuous* G-module, i.e. an abelian group with an action of G by automorphisms such that $\forall a \in A$, the stabilizer G_a is open in G (exercise: this is equivalent to the map $G \times A \to A$ being continuous, where A has the discrete topology). In particular, if $G_\alpha = G/U_\alpha$, then A^{U_α} is a G_α-module, and we define:

Definition 3.3.

$$H^i_{\mathrm{cont}}(G, A) \coloneqq \varinjlim H^i(G_\alpha, A^{U_\alpha}),$$

where the limit is taken with respect to the inflation maps induced by the projections $G_\beta \twoheadrightarrow G_\alpha$:

$$H^i(G_\alpha, A^{U_\alpha}) \to H^i(G_\beta, A^{U_\beta}).$$

We'll immediately drop the subscript cont, writing $H^i(G, A)$.

In the case where $G = \mathrm{Gal}(k^s|k)$ is the absolute Galois group of a field k, we'll use the notation $H^i(k, A)$ for $H^i(G, A)$.

Proposition 3.4. If G is profinite and A is a continuous G-module, then $H^i(G, A)$ is a torsion abelian group for all $i > 0$.

Proof. The groups $H^i(G_\alpha, A^{U_\alpha})$ are torsion: they are killed by the order of G_α. Therefore, their direct limit $H^i(G, A)$ is also torsion. □

Corollary 3.5. If moreover A is a $\mathbf{Q}$-vector space, then $H^i(G, A) = 0$ for all $i > 0$.

Remark 3.6. Given a finite group G and a finitely generated G-module A, the cohomology groups $H^i(G, A)$ are finite for all $i > 0$. This follows from the description of cohomology in terms of the standard resolution and the torsion property above.

Lemma 3.7. Let G be profinite with (U_α) a system of open normal subgroups. Then,

- If H is a closed subgroup of G, there is a canonical isomorphism
$$H^i(H, A) \cong \varinjlim H^i(H/(H \cap U_\alpha), A^{H \cap U_\alpha})$$
which shows that H is profinite;
- If moreover H is normal in G, then $G/H \cong \varprojlim G/(HU_\alpha)$ is profinite.

Therefore, we can define restriction, corestriction and inflation maps for profinite groups by taking the direct limit of the corresponding maps for finite groups: for instance, restriction is induced by the maps

$$H^i(G/U_\alpha, A^{U_\alpha}) \xrightarrow{\mathrm{res}} H^i(H/(H \cap U_\alpha), A^{H \cap U_\alpha}).$$

Since we defined cohomology explicitly as a direct limit, we have to show "by hand" the existence of the long exact sequence in cohomology associated with a short exact sequence of G-modules $0 \to A \to B \to C \to 0$.

Lemma 3.8. Let $(A_\alpha, \phi_{\alpha\beta})$, $(B_\alpha, \psi_{\alpha\beta})$ and $(C_\alpha, \rho_{\alpha\beta})$ be direct systems of G_α-modules, and assume given short exact sequences

$$0 \to A_\alpha \xrightarrow{f_\alpha} B_\alpha \xrightarrow{g_\alpha} C_\alpha \to 0$$

for all α which are compatible with the transition maps (i.e. morphisms of direct systems). Then, the direct limit of these sequences is exact.

The non-trivial part is defining the connecting homomorphism

$$\delta : H^i(G, C) \to H^{i+1}(G, A).$$

Let $U \subset G$ be an open normal subgroup. If $K_U \coloneqq \operatorname{coker}(B^U \to C^U,)$ then $\varinjlim_U K_U = 0$. By the lemma, we have a short exact sequence

$$\varinjlim H^i(G/U, (B^U/A^U)) \to \varinjlim H^i(G/U, C^U) \to \varinjlim H^{i+1}(G/U, K_U)$$

and the last term is 0 by the above. Therefore, for small enough U we can lift a cocycle $f \in Z^i(G, C)$ to some element in $\varinjlim H^i(G/U, (B^U/A^U))$, and then apply the connecting homomorphism coming from the short exact sequence $0 \to A^U \to B^U \to B^U/A^U \to 0$ to get an element in $\varinjlim H^{i+1}(G/U, A^U)$, which has a map into $H^{i+1}(G, A)$ (of course, one should check that this does not depend on any of the choices made).

4 Kummer Theory

Proposition 4.1 (Hilbert's Theorem 90)**.** Let $K|k$ be a finite Galois extension with group $G = \operatorname{Gal}(K|k)$. Then

$$H^1(G, K^\times) = 0.$$

Consequently, by passing to the limit over all such extensions we get

$$H^1(k, (k^s)^\times) = 0.$$

Lemma 4.2 (Dedekind)**.** If $\sigma_1, \ldots, \sigma_n$ are the elements of G, than they are linearly independent over K.

Proof. Suppose they are linearly dependent, and let

$$a_1\sigma_1 + \ldots + a_n\sigma_n = 0$$

be the *shortest* nontrivial linear relation. Then, choosing $x \in K$ such that $\sigma_1(x) \neq \sigma_2(x)$ and $y \in K$ is arbitrary, then

$$\begin{aligned} 0 &= a_1\sigma_1(xy) + \ldots + a_n\sigma_n(xy) \\ &= a_1\sigma_1(x)\sigma_1(y) + \ldots + a_n\sigma_n(x)\sigma_n(y) \\ &= \sigma_1(x)(a_1\sigma_1(y) + \ldots + a_n\sigma_n(y)) + (\sigma_2(x) - \sigma_1(x))(a_2\sigma_2(y) + \ldots + a_n\sigma_n(y)). \end{aligned}$$

Putting this together with the original relation evaluated at y and multiplied by $\sigma_1(x)$, we get a shorter relation

$$(\sigma_2(x) - \sigma_1(x))(a_2\sigma_2(y) + \ldots + a_n\sigma_n(y)) = 0,$$

which is a contradiction. □

Proof of Proposition. Let $\varphi : G \to K^\times$ be a 1-cocycle, with the cocycle condition being $\varphi(\sigma\tau) = \varphi(\sigma) \cdot \sigma(\varphi(\tau))$. We want to show that φ is a coboundary, i.e. there exists some $y \in K^\times$ such that $\varphi(\tau) = \tau(y) \cdot y^{-1}$ for all $\tau \in G$. Consider the element

$$\Phi = \sum_{\sigma \in G} \varphi(\sigma) \cdot \sigma \in K[G].$$

By the lemma, Φ is non-zero. Therefore there exist $x, y \in K^\times$ such that

$$y = \Phi(x) = \sum_{\sigma \in G} \varphi(\sigma) \cdot \sigma(x),$$

and applying τ to both sides gives

$$\tau(y) = \sum_{\sigma \in G} \tau\varphi(\sigma) \cdot \tau\sigma(x) = \sum_{\sigma \in G} \frac{\varphi(\tau\sigma)}{\varphi(\tau)} \tau\sigma(x) = \frac{1}{\varphi(\tau)} \sum_{\sigma \in G} \varphi(\tau\sigma) \cdot \tau\sigma(x) = \frac{1}{\varphi(\tau)} \sum_{\sigma \in G} \Phi(x) = \frac{y}{\varphi(\tau)},$$

□

The main result of this section is the following:

Proposition 4.3. Let k be a field and m a positive integer which is prime to $\mathrm{char}(k)$ (this implies that $\mu_m \subset k^s$). Then, there is a canonical isomorphism

$$H^1(k, \mu_m) \cong k^\times / k^{\times m}.$$

whose inverse sends the class of $a \in k^\times / k^{\times m}$ to the class of the cocycle $\sigma \mapsto \sigma(\alpha) \cdot \alpha^{-1}$, where $\alpha \in (k^s)^\times$ is any m-th root of a.

Proof. Consider the short exact sequence of G_k-modules

$$1 \to \mu_m \to (k^s)^\times \xrightarrow{(\cdot)^m} (k^s)^\times \to 1.$$

By Hilbert's Theorem 90, we have $H^1(k, (k^s)^\times) = 0$. Therefore, the long exact sequence in cohomology gives

$$H^0(k, (k^s)^\times) \xrightarrow{(\cdot)^m} H^0(k, (k^s)^\times) \to H^1(k, \mu_m) \to 0,$$

and the cokernel of the first map is exactly $k^\times / k^{\times m}$, since $H^0(k, (k^s)^\times) = (k^s)^{\times G_k} = k^\times$. The last statement then follows by writing out the connecting map explicitly (lifting corresponds to taking an m-th root…). □

Example 4.4. If moreover $\mu_m \subset k$, then $H^1(k, \mu_m) = H^1(k, \mathbf{Z}/m\mathbf{Z}) = \mathrm{Hom}(G_k, \mathbf{Z}/m\mathbf{Z})$ (the action is trivial). In this case, we deduce that every finite extension $K|k$ with $\mathrm{Gal}(K|k) \cong \mathbf{Z}/m\mathbf{Z}$ is of the form $K = k(\alpha)$ with $\alpha^m \in k$.

5 Cup-products

Let G be a group and A, B be G-modules. The module $A \otimes_{\mathbf{Z}} B$ is a G-module with the diagonal action $\sigma(a \otimes b) = \sigma(a) \otimes \sigma(b)$. The cup-product is a bilinear pairing

$$\cup : H^i(G, A) \times H^j(G, B) \to H^{i+j}(G, A \otimes B)$$
$$(a, b) \mapsto a \cup b$$

which is functorial in A, B and graded-commutative: $a \cup b = (-1)^{ij} b \cup a$; there will be a version for profinite groups by passing to the limit. Let us sketch the application of this gadget to the Hilbert symbol:

Definition 5.1. For a field k, the *Brauer group* of k is

$$\mathrm{Br}(k) \coloneqq H^2(k, k_s^\times).$$

A non-trivial result is that elements of $\mathrm{Br}(k)$ correspond to central division algebras which are finite-dimensional over k.

Another piece of the Kummer long exact sequence is

$$H^1(k, k_s^\times) = 0 \to H^2(k, \mu_n) \xrightarrow{(\cdot)^n} H^2(k, k_s^\times),$$

from which we get $H^2(k, \mu_n) \cong {}_n\mathrm{Br}(k)$, the n-torsion in the Brauer group.

Suppose that $\mu_n \subset k$, and take $a, b \in k^\times$. Then, we can take their classes in $k^\times / k^{\times n} \cong H^1(k, \mu_n) = H^1(k, \mathbf{Z}/n\mathbf{Z})$, and take their cup-product:

$$H^1(k, \mathbf{Z}/n\mathbf{Z}) \times H^1(k, \mathbf{Z}/n\mathbf{Z}) \to H^2(k, \mathbf{Z}/n\mathbf{Z} \otimes_{\mathbf{Z}} \mathbf{Z}/n\mathbf{Z}) \cong {}_n\mathrm{Br}(k).$$

Theorem 5.2 (Hasse)**.** Let $k|\mathbf{Q}_p$ be a finite extension. Then,

$$\mathrm{Br}(k) \cong \mathbf{Q}/\mathbf{Z}.$$

In particular, we deduce that for any n ${}_n\mathrm{Br}(k) \cong \frac{1}{n}\mathbf{Z}/\mathbf{Z}$. Putting these together, we get a map

$$(\cdot, \cdot)_n : k^\times / k^{\times n} \times k^\times / k^{\times n} \to H^1(k, \mathbf{Z}/n\mathbf{Z}) \times H^1(k, \mathbf{Z}/n\mathbf{Z}) \to {}_n\mathrm{Br}(k) \cong \mathbf{Z}/n\mathbf{Z},$$

which is the *Hilbert symbol*. What happens if $\mu_n \not\subset k$? Given a character $\chi : G_k \to \mathbf{Z}/n\mathbf{Z}$, this lives in the first cohomology group $H^1(k, \mathbf{Z}/n\mathbf{Z})$, so we can take its cup-product with the class of any $b \in k^\times$ inside $H^1(k, \mu_n)$:

$$(\chi, b) \coloneqq \chi \cup b \in H^2(k, \mu_n) \overset{\text{Hasse}}{\cong} \mathbf{Z}/n\mathbf{Z}.$$

Construction 5.3. Let $A^\bullet, B^\bullet$ be complexes of e.g. abelian groups. The tensor product of

these complexes is defined as follows: consider the double complex

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \uparrow & & \uparrow & & \uparrow & \\
A^{i-1}\otimes B^{j+1} & \longrightarrow & A^{i}\otimes B^{j+1} & \longrightarrow & A^{i+1}\otimes B^{j+1} & \longrightarrow & \dots \\
 & \uparrow & & \uparrow & & \uparrow & \\
A^{i-1}\otimes B^{j} & \longrightarrow & A^{i}\otimes B^{j} & \longrightarrow & A^{i+1}\otimes B^{j} & \longrightarrow & \dots \\
 & \uparrow & & \uparrow & & \uparrow & \\
A^{i-1}\otimes B^{j-1} & \longrightarrow & A^{i}\otimes B^{j-1} & \longrightarrow & A^{i+1}\otimes B^{j-1} & \longrightarrow & \dots
\end{array}$$

with horizontal and vertical differentials

$$\partial^h_{i,j} = \partial^i_A \otimes \mathrm{id}_{B^j}, \qquad \partial^v_{i,j} = (-1)^i \,\mathrm{id}_{A^i} \otimes \partial^j_B.$$

With this sign, the diagram anti-commutes, i.e. $\partial^h_{i,j+1} \circ \partial^v_{i,j} = -\partial^v_{i+1,j} \circ \partial^h_{i,j}$. Recall that the total complex $T(A^\bullet \otimes B^\bullet)$ in this case is defined as

$$T^n(A^\bullet \otimes B^\bullet) = \bigoplus_{i+j=n} A^i \otimes B^j,$$

with differential given on each component by the sum of the horizontal and vertical differentials:

$$\partial^n = \sum_{i+j=n} \partial^h_{i,j} + \partial^v_{i,j}.$$

Because of the anti-commutativity, we have $\partial^{n+1} \circ \partial^n = 0$, which shows that this is a complex.

Definition 5.4. We denote by $(A \otimes B)^\bullet$ the total complex $T(A^\bullet \otimes B^\bullet)$, and call it the *tensor product* of the two complexes.

Assume moreover that A, B are abelian groups, and consider the complexes $\mathrm{Hom}(A^\bullet, A)$ and $\mathrm{Hom}(B^\bullet, B)$, with differentials given by pre-composition with ∂_A, ∂_B. We want to define a product

$$H^i(\mathrm{Hom}(A^\bullet, A)) \times H^j(\mathrm{Hom}(B^\bullet, B)) \to H^{i+j}(\mathrm{Hom}((A \otimes B)^\bullet, A \otimes B)).$$

If $\alpha \in \mathrm{Hom}(A^i, A)$ and $\beta \in \mathrm{Hom}(B^j, B)$, then $\alpha \otimes \beta : A^{-i} \otimes B^{-j} \to A \otimes B$ and we get a map

$$\mathrm{Hom}(A^{-i} \otimes B^{-j}, A \otimes B) \to \mathrm{Hom}\left(\bigoplus_{k+l=i+j} A^{-k} \otimes B^{-l}, A \otimes B \right) = \mathrm{Hom}(T^{i+j}(A^\bullet \otimes B^\bullet), A \otimes B),$$

and one easily checks that this is compatible with differentials (sends cocycles to cocycles and coboundaries to coboundaries). Therefore, it induces such a product on cohomology, which is the desired cup-product.

Lemma 5.5. Let $P_\bullet \to \mathbf{Z}$ be a projective resolution of $\mathbf{Z}$ as a trivial G-module, and recall the homological indexing $P_i = P^{-i}$. Then, $(P \otimes P)_\bullet$ is a projective resolution of $\mathbf{Z}$ as a trivial $(G \times G)$-module, where the action is the diagonal one.

Applying the previous construction with $A^\bullet = B^\bullet = P_\bullet$, we get a product

$$H^i(\mathrm{Hom}_G(P_\bullet, A)) \times H^j(\mathrm{Hom}_G(P_\bullet, B)) \to H^{i+j}(\mathrm{Hom}_{G\times G}((P \otimes P)_\bullet, A \otimes B)),$$

and composing with the map $H^{i+j}(\mathrm{Hom}_{G\times G}((P\otimes P)_\bullet, A\otimes B)) \to H^{i+j}(\mathrm{Hom}_G((P\otimes P)_\bullet, A\otimes B))$ induced by the diagonal embedding $G \hookrightarrow G \times G$, we get a cup-product

$$H^i(G, A) \times H^j(G, B) \to H^{i+j}(G, A \otimes B).$$

This is compatible with restriction, inflation and corestriction maps: if $H \leq G$ is a subgroup, then

- For $a \in H^i(G, A)$ and $b \in H^j(G, B)$, we have $\mathrm{Res}(a\cup b) = \mathrm{Res}(a)\cup\mathrm{Res}(b) \in H^{i+j}(H, A\otimes B)$.
- If $H \triangleleft G$ is normal, then for $a \in H^i(G/H, A^H)$ and $b \in H^j(G/H, B^H)$, we have $\mathrm{Inf}(a\cup b) = \mathrm{Inf}(a) \cup \mathrm{Inf}(b) \in H^{i+j}(G, A \otimes B)$.
- If $[G : H] < \infty$, then for $a \in H^i(H, A)$ and $b \in H^j(G, B)$, we have

  $$\mathrm{Cor}(a \cup \mathrm{Res}(b)) = \mathrm{Cor}(a) \cup b \in H^{i+j}(G, A \otimes B)$$

  which is called the *projection formula*.

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